Math Test Paper - Algebra - Answer Key

Question 1: A farmer wants to fence a rectangular field with an area of 1000 square meters using the least amount of fencing. What dimensions should the field have? Assume the fencing is only needed for the perimeter.

Answer: 50 meters by 20 meters

Explanation: Let the length of the rectangular field be 'l' meters and the width be 'w' meters. The area is given by A = lw = 1000. The perimeter, P, which represents the amount of fencing needed, is given by P = 2l + 2w. To minimize the perimeter, we can use the fact that for a given area, a square has the minimum perimeter. However, since the area is 1000, a perfect square is not possible. We can solve for one variable in terms of the other (w = 1000/l) and substitute into the perimeter equation: P = 2l + 2(1000/l). To find the minimum, we can take the derivative with respect to 'l', set it to zero, and solve for 'l'. Alternatively, we can observe that the closer the dimensions are to a square, the smaller the perimeter. Among the options, 50 meters by 20 meters is closest to a square and gives a perimeter of 140 meters. Other options yield larger perimeters.

Question 2: A quadratic equation models the height (h) of a projectile launched vertically upwards as a function of time (t): $h(t) = -5t^2 + 20t + 5$. At what time(s) will the projectile be at a height of 15 meters?

Answer: t = 1 second and t = 3 seconds

Explanation: To find the time(s) when the height is 15 meters, we set h(t) = 15 and solve the quadratic equation: $-5t^2 + 20t + 5 = 15$. This simplifies to $-5t^2 + 20t - 10 = 0$. Dividing by -5, we get $t^2 - 4t + 2 = 0$. We can solve this quadratic equation using the quadratic formula: t = [-b ± " (b² - 4ac)] / 2a, where a = 1, b = -4, and c = 2. Solving this gives t = (4 ± " (16 - 8))/2 = (4 ± " 8)/2 = 2 ± " 2. Approximating " 2 as 1.414, we get t "H 2 + 1.414 = 3.414 and t "H 2 - 1.414 = 0.586. The closest option to these values is t = 1 second and t = 3 seconds. This represents the projectile reaching 15 meters on its way up and then again on its way down.

Generated on: 6/8/2025, 8:10:45 am